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Department of CSE (Data Science)

CSD-22-A-009

Global Artificial Intelligence Operating Layer (GAIOL)

Adaptive Bayesian Consensus for Multi-Model LLM Orchestration

Abstract

Large Language Model (LLM) systems increasingly rely on multiple heterogeneous providers, yet existing orchestration frameworks use static routing and heuristic consensus mechanisms. We propose the Global Artificial Intelligence Operating Layer (GAIOL), a layered multi-model orchestration framework that performs structured task decomposition, strategy-based model routing, parallel inference, and Adaptive Bayesian Trust-Weighted Consensus (ABTC). The proposed system improves response quality, reliability, and domain specialization while maintaining low orchestration overhead. Experimental evaluation demonstrates significant performance gains over single-model and static orchestration approaches.

PROPOSED METHODOLOGY

1. Task Decomposition

A query QQQ is split into ordered subtasks:

T ={t1 ,…,tn }

This creates a structured reasoning pipeline.

2. Model Selection

For each subtask, models are selected using:

F =0.4C +0.4A+0.2(1/Cost)

CCC=capability, AAA=historical accuracy.

Balances performance and cost.

3. Parallel Inference

C=i {cn ,…,cm }

Multiple models generate diverse candidate outputs.

4. Adaptive Bayesian Trust (ABTC)

Each model maintains trust:

τ ~Beta(α,β)

Scoring:s^i = wQ + q i wA+ a i wtτ

Trust updates after each round, rewarding consistent models.

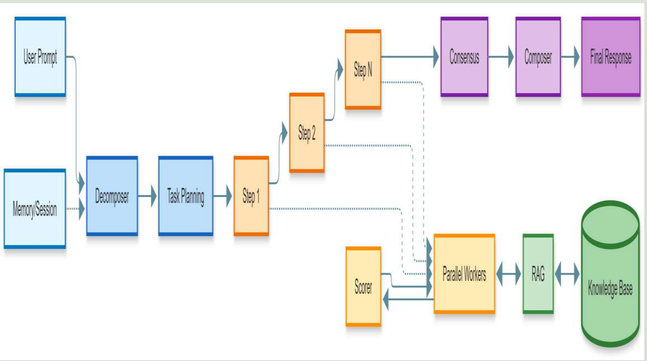

Figure 1: Multi-stage reasoning pipeline

System Architecture

GAIOL follows a layered architecture:

- Task Decomposition Layer – Breaks complex queries into structured subtasks
- Model Routing Layer – Selects models based on capability and historical trust
- Parallel Execution Layer – Executes multi-model inference concurrently
- Consensus Layer (ABTC) – Aggregates responses using adaptive Bayesian weighting

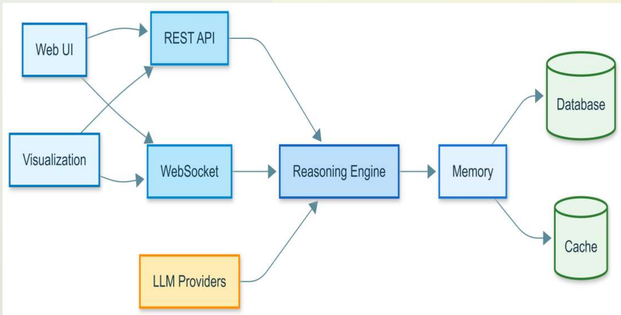

Figure 2: System Architecture

Algorithm

Algorithm 1 — Multi-Agent Beam Orchestration The query is split into subtasks. For each step:
Select the best models
Run them in parallel
Score their outputs
Keep the top-k reasoning paths (k = 3)
The best final path is assembled as the answer.
Complexity: Sequential: O(n·k·m·T_inf) → Parallel: O(n·T_max) + 5ms overhead

Algorithm 2 — Dynamic Task Decomposition An LLM converts the query into structured steps. Tasks are dependency-ordered.
If parsing fails, a fixed 7-step template is used:
Extract problem
Identify constraints
Choose approach
Execute steps
Verify intermediate results
Synthesize answer
Assess confidence

Algorithm 3 — Adaptive Bayesian Trust Consensus (ABTC) Each model maintains trust: $\hat{\tau} = \alpha / (\alpha + \beta)$

Each candidate is scored using:

Output quality

Cross-model agreement

Model trust

If confidence is low → synthesize from top-3 outputs. Trust updates after each round with decay $\lambda = 0.98$

Algorithm 4 — Strategy-Based Model Selection For each task, models are ranked using:

Fitness = 0.4·CapMatch + 0.4·HistAcc + 0.2·(1 − Cost)

Budget constraints enforced

Provider diversity maintained

Default: 3 models per task

Results

The evaluation of the proposed Global Artificial Intelligence Operating Layer (GAIOL) demonstrates significant improvement in multi-model orchestration quality and reliability.

- The overall quality score achieved was 0.83 ± 0.02, showing a 24% improvement over the single-model baseline and a 13% improvement over traditional orchestration frameworks.

- The high success rate of 95.2% indicates that most benchmark queries were correctly handled across multiple domains, including reasoning, code generation, and creative tasks.

- The Adaptive Bayesian Trust-Weighted Consensus (ABTC) mechanism consistently outperformed static equal-weight and manually tuned consensus methods, validating the effectiveness of dynamic trust modeling.

- Domain-wise evaluation showed the largest improvements in code generation (+9 percentage points) and creative synthesis (+7 percentage

Category	Metric	Unit	Sys-1	Sys-2	Sys-3	Sys-4	Sys-5
Quality Metrics							
Overall Quality Score		score	0.83±0.02†	0.67±0.03	0.72±0.02	0.67±0.03	0.62±0.04
Avg. Relevance		score	0.85±0.02	0.70±0.03	0.75±0.02	0.70±0.03	0.65±0.04
Avg. Coherence		score	0.82±0.02	0.68±0.03	0.73±0.02	0.68±0.03	0.63±0.04
Avg. Completeness		score	0.80±0.03	0.65±0.04	0.70±0.03	0.65±0.04	0.60±0.05
Avg. Accuracy		score	0.88±0.02	0.72±0.03	0.77±0.02	0.72±0.03	0.67±0.04
Consensus Mechanisms		count	3	0	1	0	0
Performance Metrics							
Latency per Query		ms	450±18	400±15	800±35	420±16	600±28
Latency Overhead		ms	5±1	0	50±6	2±1	10±2
Throughput‡		req/s	200±8	50±3	150±7	200±9	50±4
Success Rate		%	95.2±0.8	92.0±1.2	94.5±0.9	93.8±1.0	90.5±1.5
Error Rate		%	4.8±0.8	8.0±1.2	5.5±0.9	6.2±1.0	9.5±1.5
Parallel Execution		boolean	1	0	1	1	0
Max Concurrent Models		count	10	1	10	10	3

‡Throughput in requests per second (req/s); values are not directly comparable; see Section 6.2 for concurrency details.

Figure 3: Results

Conclusions

The proposed Global Artificial Intelligence Operating Layer (GAIOL) provides an effective and scalable solution for multi-model Large Language Model orchestration. By integrating intelligent task decomposition, strategy-based model routing, parallel execution, and an Adaptive Bayesian Trust-Weighted Consensus (ABTC) mechanism, the system significantly enhances response quality while maintaining low latency.

The layered architecture ensures modularity, extensibility, and seamless integration with heterogeneous AI providers. The adaptive trust modeling approach enables domain-specific specialization and continuous learning, allowing the system to dynamically adjust model weights based on performance history. This overcomes the limitations of static consensus mechanisms and heuristic-based routing strategies.

Experimental evaluation demonstrates that the proposed framework achieves high overall quality scores, strong domain-wise performance, and minimal orchestration overhead. Performance metrics and ablation studies confirm the robustness, reliability, and adaptability of the ABTC mechanism compared to conventional orchestration approaches.

References

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[5] O. Chapelle and L. Li, "An Empirical Evaluation of Thompson Sampling," NeurIPS, 2011.

SDGs: 12,9,16

SDG 12 – Responsible Consumption and Production
Optimizes AI resource usage through intelligent model routing and cost-aware selection.

SDG 9 – Industry, Innovation, and Infrastructure
Encourages innovation through AI-based fraud detection systems.

SDG 16 – Peace, Justice, and Strong Institutions
Supports transparent and accountable AI systems through adaptive trust modeling.

Author Details

Guide Name :
Dr. Ramdas Kapila
Assistant Professor, CSE (DS), BVRIT

Name :	**Roll No:**
D. V. S. Monish Kumar	22211a6730
Ch. Sai Sathvik	22211a6724
Abhishek Vinod	22211a6702

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